BIOCHEMISTRY CUMULATIVE EXAM

April 26, 1997

A hemoglobin molecule has been discovered in a Columbian mosquito that consists of one subunit and binds a single molecule of O_2 to the iron. It can also bind a single molecule of CO_2 to the α amino group. The binding of O_2 to hemoglobin is described by a graph of y vs. pO_2, where y is the fractional saturation ($y = [Hb\text{-}O_2]/[\text{total Hb}]$), and pO_2 is the partial pressure of O_2. p_{50} is a constant which is equal to the value of pO_2 required to give $y = 0.5$. In the absence of CO_2, the O_2 binding curve is hyperbolic with $p_{50} = 6$ torr. When the CO_2 pressure is increased, the value of p_{50} for O_2 increases until it reaches a plateau value of $p_{50} = 60$ torr when the CO_2 pressure is greater than 100 torr. In the absence of O_2, CO_2 binds to hemoglobin with a hyperbolic binding curve, and a p_{50} for CO_2 of 3 torr.

1. (30) Propose a simple model consisting of a set of binding equilibria showing the binding of CO_2 and O_2 to hemoglobin, which accounts for the effect of CO_2 binding on the affinity of O_2 binding.

2. (25) From your model, derive a complete equation for the fractional saturation y of O_2 bound to hemoglobin as a function of pO_2 and pCO_2.

3. (25) Calculate the values for all of the constants in your equation from the above data. Calculate the value of p_{50} for CO_2 when the partial pressure of O_2 is very large, greater than 200 torr.

4. (20) Discuss why this binding behavior is beneficial for hemoglobin function in the mosquito.

Biochemistry Cumulative Examination

The following is excerpted from a New England Biolabs advertisement. You will be asked questions about the ad copy:

The LITMUS plasmids (**L**ogical **I**n vitro **T**ranscription, **M**ultiple **U**nique **S**ites) are a set of 4 multipurpose cloning/in vitro transcription phagemid vectors from New England Biolabs. These vectors feature a large number of unique polylinker cloning sites, blue-white screening, the high-copy pUC replication origin, an M13 origin for production of single-stranded mutagenesis template, and a novel system for generating RNA probes from either insert strand using only T7 RNA polymerase.

1. What are "polylinker cloning sites?" How are they useful? (10 points)

The LITMUS plasmids have the same origin of replication and high copy number as the familiar pUC vectors.

2. What is a replication origin? What is meant by "high-copy....replication origin?" Why is this good? (15 points)

Blue/White Selection

The LITMUS polylinkers are located in the *lacZ'* gene: disruption of the *lacZ* alpha-peptide by cloning foreign DNA destroys alpha-complementation in *E. coli* strains harboring the lacZ M15 deletion. LacZ delM15 strains containing the LITMUS plasmids form blue colonies on X-Gal plates; when a fragment is inserted in the polylinker, the colonies are white.

3. X-Gal is a dye linked to galactose which is readily hydrolyzed by β-galactosidase. When hydrolyzed the dye is blue, when linked to the galactose it is colorless. Given this information what is "alpha-complementation?" Explain the utility of "blue/white selection." (20 points)

M13 Origin of Replication

The LITMUS plasmids contain the origin of replication from the filamentous coliphage M13, in a single orientation. Single-stranded template DNA for in vitro mutagenesis and DNA sequencing can be produced in high yields by superinfecting with an M13 helper phage (e.g. M13KO7).

4. What is an "M13 helper phage" and how does it interact with "the origin of replication from the filamentous coliphage M13" to produce single stranded DNA? (10 points)

Linker Addition

The LITMUS polylinkers all contain a centrally located *Eco*R V site that can be used for adding sticky ends to blunt-ended DNA fragments. *Eco*R V leaves blunt ends....

5. What is the difference between "blunt ends" and "sticky ends?" (20 points)

Nested deletions

Sites for enzymes that yield a 3' overhang upon cleavage (*Pst* I, *Nsi* I, *Aat* II, *Sac* I, *Kpn* I, *Apa* I and *Sph* I) have been evenly distributed within the LITMUS polylinkers. Since 3' overhangs are resistant to the action of exonuclease III, this arrangement allows generation of unidirectional nested deletions of a cloned insert using NEB's Exo-Size Deletion Kit. This permits sequencing of a large insert using a single sequencing primer, without time-consuming "primer walking."

6. What does an exonuclease do? Reasoning from the information above and with the added information that exonuclease III does not act at 5' ends, does exonuclease III act upon single stranded or duplex DNA? (25 points)